USN | | | | | | | | | | [22CSE156]

FIFTH SEMESTER B. Tech EXAMINATION, DECEMBER - 2024

ESSENTIALS OF MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions: **I.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Discuss machine learning with its importance.	L2	CO1	**02**
	b.	Differentiate linear and logistic regression.	L3	CO2	**02**
	c.	Illustrate the concept of Principal Component Analysis (PCA).	L2	CO3	**02**
	d.	Interpret the role of entropy in decision tree learning.	L2	CO4	**02**
	e.	Outline the K-Means clustering and its main objective.	L2	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain the different types of learning in machine learning.	L2	CO1	**05**
		OR			
	b.	Summarize the role of exploratory data analysis (EDA) in machine learning.	L2	CO1	**05**
3.	**a.**	Describe how Ordinary Least Squares (OLS) is used in linear regression?	L2	CO2	**05**
		OR			
	b.	Compare logistic regression with Naive Bayes classification.	L2	CO2	**05**
4.	**a.**	Illustrate the different types of kernels used in support vector machines (SVM).	L2	CO2	**05**
		OR			
	b.	Sketch the concept of confusion matrix and its significance in classification models.	L3	CO2	**05**
5.	**a.**	Differentiate the hyperparameter tuning and its importance in model optimization.	L4	CO3	**05**
		OR			
	b.	Contrast the Principal Component Analysis (PCA) and the Linear Discriminant Analysis (LDA).	L4	CO3	**05**
6.	**a.**	Interpret the ID-3 Algorithm in decision tree learning and its key steps.	L2	CO4	**05**
		OR			
	b.	Illustrate the key issues in decision tree learning.	L2	CO4	**05**

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7.	**a.**	How does hierarchical clustering differ from K-Means clustering?	L3	CO5	**05**
		OR			
	b.	Demonstrate the difference between reinforcement learning and supervised learning?	L2	CO6	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Describe the process of designing a machine learning system with an example.	L2	CO1	**10**
		OR			
	b.	Discuss the role of NumPy and Pandas in data preprocessing and analysis for machine learning.	L2	CO1	**10**
9.	**a.**	Employ the working of logistic regression and how it can be used for binary classification.	L3	CO2	**10**
		OR			
	b.	Illustrate the performance of a linear machine learning model using confusion matrix, accuracy, precision, and recall.	L3	CO2	**10**
10.	**a.**	Discuss the role of hyperplane and decision surface in support vector machines (SVM).	L2	CO2	**10**
		OR			
	b.	Explain the concept of Maximum Likelihood Estimation (MLE) and its application in parameter estimation.	L2	CO3	**10**
11.	**a.**	Demonstrate how Principal Component Analysis (PCA) is used for dimension reduction and how it improves model performance.	L3	CO3	**10**
		OR			
	b.	Dramatize the entropy and information gain concepts in decision tree learning with an example.	L3	CO4	**10**
12.	**a.**	Differentiate the concept of Random Forest and how it enhances decision tree learning.	L4	CO4	**10**
		OR			
	b.	Compare and contrast K-Means clustering and DBSCAN in terms of algorithm and real-world applications.	L4	CO5	**10**
13.	**a.**	Dramatize the reinforcement learning and describe the methods used to teach a system to maximize rewards.	L3	CO6	**10**
		OR			
	b.	Demonstrate how does fixing the value of K affects the performance of K-Means clustering? Explain with examples.	L3	CO5	**10**